

Weekly Progress Report - Week 02

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Domain: Machine Learning
Project Name: Smart City Traffic Forecasting
Week Ending: 02
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I. Overview

During Week 2, the focus was on feature engineering, machine learning model development, and performance evaluation for the Smart City Traffic Forecasting project. Multiple regression models were trained and compared to identify the most suitable model for forecasting traffic volumes.

II. Achievements

1. Feature Engineering

- Extracted temporal features including Year, Month, Day, Hour, and Day of Week from the DateTime column.
- Prepared the dataset for machine learning model training.

2. Model Development

- Implemented Linear Regression.
- Implemented Decision Tree Regressor.
- Implemented Random Forest Regressor.

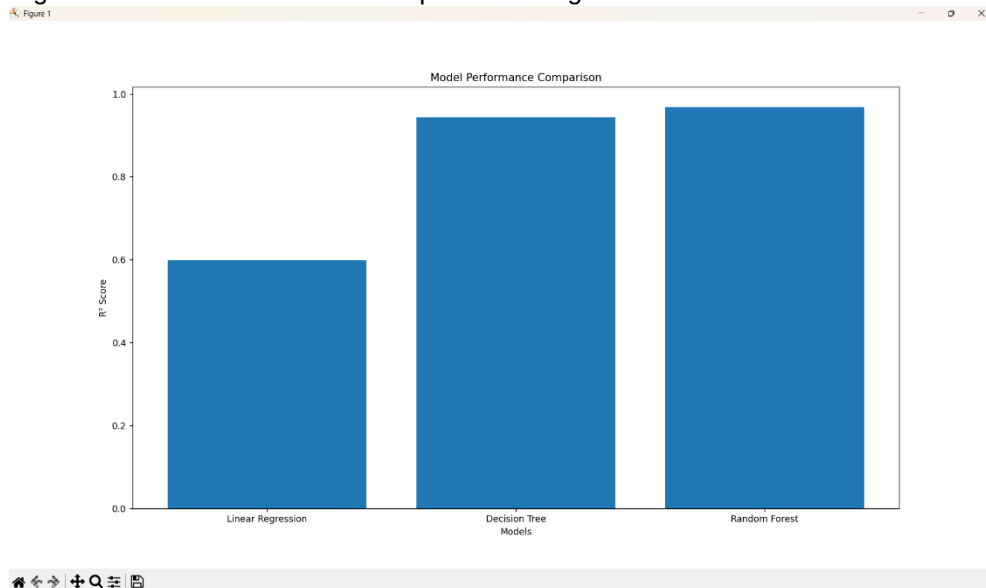
3. Model Evaluation

- Evaluated models using MAE, RMSE, and R^2 Score.
- Compared model performances using graphical visualization.

4. Results

- Linear Regression R^2 Score: 0.5997
- Decision Tree R^2 Score: 0.9436
- Random Forest R^2 Score: 0.9690
- Random Forest achieved the highest prediction accuracy and was selected as the best-performing model.

Figure 1: Model Performance Comparison Using R^2 Score



III. Challenges

- Understanding differences in model performance.
- Selecting appropriate evaluation metrics for regression problems.
- Optimizing model accuracy while maintaining computational efficiency.

IV. Learning Resources

- Scikit-Learn Documentation
- Random Forest Regressor Documentation
- Decision Tree Regressor Documentation
- Machine Learning Regression Tutorials

V. Next Week Goals

- Perform hyperparameter tuning on the Random Forest model.
- Improve forecasting accuracy.
- Generate prediction visualizations.
- Prepare the model for final testing and reporting.

VI. Additional Comments

Week 2 successfully established a predictive modeling pipeline. Random Forest demonstrated the best performance with an R^2 score of 0.9690 and was selected for future forecasting tasks.